MACHINE LEARNING PROJECT / TASK 1: PROJECT PROPOSAL

Detecting Fraudulent Financial Transactions Using Machine Learning

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1. Dataset Description

The study will use the **Credit Card Fraud Detection** dataset published by the Machine Learning Group of Université Libre de Bruxelles (ULB) through Kaggle [1]. It contains **284,807** European credit-card transactions recorded over two days in September 2013, including **492 fraudulent cases (0.172%)**. The 30 numerical predictors comprise **Time** (elapsed seconds), **Amount** (transaction value), and **V1–V28** (anonymized principal components). The target **Class** is labelled 0 = legitimate and 1 = fraudulent. Kaggle lists Open Database/Database Contents licence terms (ODbL/DbCL); attribution and applicable redistribution conditions will be observed. The original feature meanings and label-verification procedure are not disclosed.

2. Problem Statement

Financial institutions process large transaction volumes, making manual examination of every payment impractical. The project asks: Can recorded transaction features identify potentially fraudulent credit-card payments for prioritised investigation? This is a supervised binary classification problem because historical transactions have legitimate/fraud labels. Intended beneficiaries include fraud analysts, payment providers and affected customers. A model would assign a fraud risk score and flag suspicious transactions for human review, not declare fraud conclusively or automatically block payments.

3. Chosen Evaluation Metric

The primary metric will be average precision (AP), a summary of the precision–recall curve that evaluates how effectively the model ranks rare fraud cases [2]. Overall accuracy is misleading when only 0.172% of records are fraudulent. Missing fraud may expose customers to losses, while false alerts consume investigators' time; therefore, fraud-class precision, recall, F1-score and the confusion matrix will be reported as supporting measures. Scores will be compared on identical validation folds.

4. Baseline Idea and Evaluation Plan

A naive majority-class baseline will predict legitimate (Class = 0) for every transaction; a constant fraud-probability score equal to the training prevalence will serve as its AP benchmark. Its fraud-class F1-score is mathematically zero when fraud cases are present; any numerical AP result must be measured on validation data rather than invented. Consistent with the earlier group activity, Logistic Regression will be a simple learned comparator and Random Forest the principal candidate. An approximately 80/20 stratified train/test split and stratified cross-validation within training data are proposed. All preprocessing and any resampling will be learned from training folds only; the test set will remain untouched until the final evaluation. A chronological split will also be assessed for deployment realism.

Ethics, Limits and Data Handling

Anonymised PCA variables limit interpretability; the source represents only two historical days. The study will review false alerts, missed fraud and potential data leakage, preserve human oversight, and avoid claiming real-world deployment readiness.

References

[1] Machine Learning Group – ULB, "Credit Card Fraud Detection," Kaggle. https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud

[2] T. Saito and M. Rehmsmeier, "The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets," PLOS ONE, vol. 10, no. 3, e0118432, 2015. https://doi.org/10.1371/journal.pone.0118432